Default Equation Mapping
Supporting document `IndigoCarbon_US-1_2023_0074`

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1 Context

This document describes how each input to the default equations was derived using Grower Survey answers and SEP parameters. (See MP Table 6.2 Biogeochemical model input variables and source for a list of all inputs, `IndigoCarbon_US-1_2023_0029` for the list of Grower Survey questions, and `SEP-Parameters-v-1.1a` for provided parameters).

Only equations that have relevant parameters coming from other sources are included in this document. For example, Equation 5.7 (Methane Emission Reductions) is not included, because this equation is only an aggregation of previous equations which do rely on data from other sources.

Default equations are calculated at the management zone/cultivation cycle level. Descriptions of all levels can be found at `IndigoCarbon_US-1_2023_0066` Data Flow Diagram.

2 Methane Emissions

2.1 Equation 5.9 Methane Emissions from the Soil Organic Carbon Pool

Not included in this issuance. See `IndigoCarbon_US-1_2023_0010` – De-minimis assessments for details on why CH_4 from upland soils are not included in this project. Our de-minimis assessment assumed all upland crop production soils were non-wetland, and wetland soils were excluded from the project. Thus, all project soils are assumed to be upland.

2.2 Equation 5.11b: Calculated Methane Emissions from Manure Deposition

- Animal grazing days ($AGD_{l,s,t}$) is calculated using the following equation:

  $stocking_rate_heads_per_ha * area_ha * grazing_hours_per_day * (event_end - event_start + 1) * hour_to_day_conversion$

  - The terms *grazing_hours_per_day*, *event_start*, and event_end come directly from grower entered data.
  - Area_ha is the area of the management zone entered by growers converted from acres to ha.
  - *Stocking_rate_heads_per_ha* is calculated by dividing the *animal_count* (provided by grower) by *area_ha*.
  - $B0, l$ and VSl are obtained from the `SEP-Parameters-v-1.0a` based on grower-reported animal types that are mapped onto SEP animal types using the mappings in Table 1.
- $Mean_grazing_temp_f$ is used to calculate the methane conversion factor (MCF_{PRP}) and density ($pCH4$). This temperature is the average between daily high and low temperature averaged overall across all days from *event_start* to *event_end*. Daily high and low temperatures come from the North American Land Data Assimilation System ($NLDAS$) publicly available dataset.
- Global warming potential for methane (GWP_{CH4}) is set to the value in the *4th IPCC Assessment Report* (AR4) for vintages (i.e., years) before 2021 and to the value from the 5th IPCC Assessment Report (AR5) for vintages on or after 2021. (See `IndigoCarbon_US-1_2022_0067a`, item 2022.A.27.). The conversion from kg to metric ton is from Equation 5.11.

2.3 Equation 5.13b: Calculated methane emissions from enteric fermentation.

- See section on equation 5.11b for $AGD_{l,s,t}$ calculation.
- Project emission factor for enteric methane emissions ($PEF_{ent,i}$) is taken from `SEP-Parameters-v-1.1a` and mapped to each field based on species of grazing animal and state where a field is located (both provided from grower survey). See table in 5.11b for mappings of grower-reported animal types to SEP animal types, and the "Grazing" tab of `SEP-Parameters-v-1.1a` for full mapping of all species.
- GWP_{CH4} and conversion from kg to ton are provided by the SEP (Table 5.1 and Equation 5.11).

2.4 Equation 5.15. Methane Emissions from Biomass Burning.

- Mass of agricultural residues burned ($MB_{c,s,t}$) is calculated by multiplying the yield in kg/ha (provided by the grower survey) with the percent residue left on the field (calculated from harvest method, see table 2.1 in `IndigoCarbon_US-1_2023_0030` Model Input Mapping for mapping of "How was this crop harvested?" to StrawStoverHayRemoval in percent). For example, a management zone with 12,000 kg/ha of corn and 90% residue left on the field would have 10,800 kg/ha of residue burned if they answered "Yes" to burning.
  - We start with a per acre term rather than total yield and dividing later as written in the SEP to use direct data from the grower survey. We convert from acres to hectares using the conversion factor of 2.47 acres to one hectare.

Table 1: Animal type index

SEP Grazing Animal Type	Grower-reported Animal Type
Dairy Cows	DAIRY_DRY_COW DAIRY_LACTATING_COWS OTHER
Dairy Heifers	DAIRY_HEIFER_REPLACEMENTS
Bulls	BULLS
Beef Cows	BEEF_MATURE_COWS_WITH_COW_CALF FEEDLOT_CATTLE
Beef Heifers	BEEF_HEIFER_REPLACEMENTS BEEF_HEIFER_STOCKERS
Steers	BEEF_STEER_STOCKERS
Bison	AMERICAN_BISON
Goats	GOATS
Sheep	FEEDER_SHEEP FLOCK_SHEEP
Horeses	HORSE
Swine average for all weight categories	GESTATING_SOWS LACTATING_SOWS GROW_FINISH_PIGS WEANING_PIGS
Poultry (using values from turkeys)	TURKEYS DUCKS LAYING_HENS BROILERS

Table 2: Residue Combustion Factors

SEP Crop Name	Grower Survey Crop Name	Combustion Factor
Wheat residues	Wheat (Spring or Winter)	0.90
Maize residues	Corn	0.80
Rice residues	Rice	0.80
Other Crop	All Other Crop Names	0.85

- Combustion factors for agricultural residue (CF_c) are taken from `SEP-Parameters-v-1.1a` and mapped to each field based on crop type (provided from grower survey). All possible cash crops are mapped to an SEP subcategory (table 2).
- The methane emission factor for the burning of agricultural residue (EFc, CH_4) is taken from the "Biomass Burning" tab of `SEP-Parameters-v-1.1a` and is the same for all crops (2.7 gCH_4/kg dry matter burnt).
- GW_{CH4} and conversion from g to ton are provided by the SEP (Table 5.1 and Equation 5.11).

3 Nitrous Oxide Emissions

3.1 Equation 5.20. Direct Nitrous Oxide Emissions from Fertilizer

- The default equation for direct N_2O emissions from synthetic fertilizer uses a total synthetic N fertilizer application rate, which is calculated using NApplicationType and NApplicationAmount (mass of N- containing synthetic fertilizer applied ($M_{SF,s,t}$) in kg/ha, and N content of synthetic fertilizer (NC_{SF}). See `IndigoCarbon_US-1_2022_0030` - Model input mapping for details on this calculation. Similarly, organic amendment rate OMADAmount ($M_{OF,s,t}$) and type (OMADType) are from the Grower Survey and % total N (OMADPercentN) is from COMET-Planner. See `IndigoCarbon_US-1_2022_0030` - Model input mapping for details.

- Emission factors for nitrous oxide emissions (EFNdirect) come from the `SEP-Parameters-v-1.0a` and are mapped based on wet/dry climates
  - Wet climates were defined as locations in either a "wet" or "moist" climate zone per the *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.*
  - Dry climates were defined as locations in a "dry" climate zone per the 2019 Refinement to the *2006 IPCC Guidelines for National Greenhouse Gas Inventories.*
  - Daily temperature data in degrees Celsius is also taken from the North American Land Data Assimilation System (NLDAS)
- Molar mass ratio of N2O to N is provided by the SEP (Table 5.1 and Equation 5.20) and the global warming potentials for N_2O (GWP_{N2O}) are obtained from the *4th IPCC Assessment Report* (AR4) for vintages (i.e., years) before 2021 and the 5th IPCC Assessment Report (AR5) for vintages (i.e., years) on or after 2021. (See `IndigoCarbon_US-1_2022_0067a`, item 2022.A.27.) The year when a cultivation cycle ends is used to determine the GWP values for that cultivation cycle.

3.2 Equation 5.21. Indirect Nitrous Oxide Emissions from Fertilizer.

- See section on Equation 5.20 for calculation of $M_{SF,s,t}$, NC_{SF}, $M_{OF,s,t}$, and NC_{OF}.
- Fractions of all synthetic N added to soils that volatilizes as NH_3 and NO_x ($FRAC_{GASF}$) are provided by SEP-Parameters-v-1.0a and are mapped using the table in the Nitrogen Factors tab. See `IndigoCarbon_US-1_2023_0030` Model input mapping for details on determining fertilizer type from grower survey data. Notably, every fertilizer type is broken down into only "Urea" and "Nitrate-based" components.
- Fraction of all organic N added to soils that volatilizes as NH_3 and NO_x ($FRAC_{GASMD}$) is provided by `SEP-Parameters-v-1.1a` and is the same for all environments and organic fertilizer types (0.21 no units).
- Emission factor for nitrous oxide emissions from atmospheric deposition (EF_{Nvolat}) is provided by `SEP-Parameters-v-1.1a` in the "Nitrogen Factors" tab and is mapped based on wet/dry climates (see Equation 5.20 section for details on climate determination).
- Fraction of N added (synthetic or organic) to soils that is lost through leaching and runoff ($Frac_{LEACH}$) is assumed to be 0 in dry climates without any irrigation. Otherwise $Frac_{LEACH}$ is 0.24 in wet climates and in dry climates that had irrigation as defined in the section for Equation 5.21 is used. Note that this follows what is written directly in Equation 5.21 of the SEP but differs from `SEP-Parameters-v-1.0a`, which indicates that all climates should have a value of 0.24.
- Emission factor for nitrous oxide emissions from leaching and runoff (EF_Nleach) is provided by SEP-Parameters-v-1.0a and is the same for all environments (0.011 no units).
- Molar mass ratio of N_2O to N and GWP_{N2O} are both provided by the SEP (Table 5.1 and Equation 5.21).

3.3 Equation 5.23. Direct Nitrous Oxide Emissions from Manure Deposition.

- See section on Equation 5.11b for $AGD_{l,s,t}$ calculation.
- Nitrogen excreted by grazing animals (N_{EXl}) is taken from the "Grazing" tab in SEP-Parameters-v-1.0a and mapped to each field based on species of grazing animal and state where a field is located (both provided directly from the grower survey). See table in 5.11b for mappings of grower-reported animal types to SEP animal types. Emission factor for nitrous oxide from manure and urine deposited on soils ($EF_{N2O,md,l}$) is provided by `SEP-Parameters-v-1.1a` and mapped based on animal type and wet/dry climates (see Equation 5.20 section for details on climate determination).
- Molar mass ratio of N_2O to N, GWP_{N2O}, and conversion from kg to ton are provided by the SEP (Table 5.1 and Equation 5.23).

3.4 Equation 5.24 . Indirect Nitrous Oxide Emissions from Manure Deposition

- See section on equation 5.11b for $AGD_{l,s,t}$ calculation.

- See section on Equation 5.23 for details on N_{EXl}.
- Fraction of manure N added to soils that volatilizes as NH_3 and NO_x ($Frac_{GASMD}$) is provided by `SEP-Parameters-v-1.1a` and is the same for all climates and manure types (0.21 no units).
- See section for Equation 5.21 for details on EF_{Nvolat}.
- Fraction of manure N added to soils that is lost through leaching and runoff ($Frac_{LEACHMD}$) is assumed to be 0 in dry climates and 0.24 in wet climates or cultivation cycles with an irrigation event.
- Emission factor for nitrous oxide emissions from leaching and runoff Is provided by `SEP-Parameters-v-1.1a` and is the same for all environments (0.011 no units).
- Molar mass ratio of N_2O to N and GWP_{N2O} are both provided by the SEP (Table 5.1 and Equation 5.24).

3.5 Equation 5.25. Nitrous Oxide Emissions from the Incorporation of All Crop Residues.

- Annual dry matter (MBg,s,t) returned to the soil is calculated for the aboveground and below ground components of each crop type:
  - Above ground annual dry matter is calculated according to the following equations:

$$- \ MB_{g,s,t} = AG_g x (1 - Frac_{Remove(g)} - ((1 - Frac_{Remove(g)}) x Frac_{Burnt(g)} x CF_g))$$

$$- \ AG - g = YieldDM_g x RAG(T)g$$

$$- \ YieldDM_g = YieldFreshg x DRYg$$

Where:

- $MB_{g,s,t}$ is the annual dry matter returned to the soil for crop species g in kg/ha. The term g is further defined as either the above ground or below ground component of an individual crop species g.
- AG_g is the above ground residue dry matter for crop species g not including any post-harvest residue removal, prior to return to soil in kg/ha
- $Frac_{Remove(g)}$ is the fraction of above ground residue removed for crop species g based on the harvest method (see Table 1). The harvest method is provided by the Grower Survey (see `IndigoCarbon_US-1_2023_0029` for details).
- $Frac_{Burnt(g)}$ is the fraction of above ground residue burnt for crop species g; $Frac_{Burnt(g)}$ is equal to 1 when the Grower Survey indicates residue was burned
- CF_g is the combustion factor of above ground residue for species g, see table included with Equation 5.15
- $YieldDM_g$ is the harvested dry matter yield for crop species g in kg/ha
- $RAG(T)_g$ is the ratio of above ground residue dry matter to harvested yield for crop species g, derived from `SEP-Parameters-v-1.1a` (see Table 2 for mapping between SEP crop and Grower Survey Crop Name)
- $YieldFresh_g$ is the harvested yield for species g derived from the Grower Survey in kg/ha. When the species is a cover crop, $YieldDM_g$ is not used. In cases where the grower-provided yield was out-of-range (see `IndigoCarbon_US-1_2022_0032`), $YieldFreshg$ was derived from the National Agricultural Statistics Service (see `IndigoCarbon_US-1_2022_0055` for details on NASS). The NASS yield value was determined using the approach described in the Hierarchy of T yields selection in `IndigoCarbon_US-1_2022_0071`.
- DRY_g is the dry matter fraction of harvested crop species g, derived from `SEP-Parameters-v-1.1a` (see Table 2 for mapping between SEP crop and Grower Survey Crop Name)
- When species g is a cover crop, $MB_{g,s,t}$ is derived directly from Ruis et al. (2019) and is equal to the cover crop biomass production as provided in Table 3, Table 4.

Table 3: Fraction of above ground residue removed for all crops based on harvest method.

Harvest Method	**Fraction of Above ground Residue Removal**
Grain Only	0
Grain with Residue Removed	0.50
Baled Straw	0.30
Silage	0.90
Haylage	0.80
Hay	0.80
Other	0
Cotton Picker or Striped	0
Digging	0

- The below ground dry matter ($MB_{g,s,t}$) is derived by multiplying the above ground dry matter and the appropriate root to shoot ratio (RS_g). For cash crops with harvested yield, the harvested dry matter yield for crop species g ($YieldDM_g$) is added to the above ground dry matter.

$$MB_{g,s,t} = (AG_g + YieldDM_g) \times RS_g$$

Where:

- RS_g is the root to shoot ratio for a crop species g as provided in `SEP-Parameters-v-1.1a` (see Table 2 for mapping between SEP crop and Grower Survey Crop Name). For cover crops, mapping of the Ruis et al. (2019) cover crop class to the Grower Survey CropName is provided in Table 4.
- For tuber crops and cover crops, $YieldDM_g = 0$ because the assumption is that there is no above ground dry matter removed during harvest.

Table 4: Index Between SEP Crop Type and Grower Survey CropName

Grower Survey CropName	**(SEP-Parameters-v1.1a)**
Alfalfa	Alfalfa
Barley	Barley
Beans/Peas	Beans & pulses
Canola	Generic value for crops not indicated below
Corn	Maize
Cotton	Generic value for crops not indicated below
Flax	Generic value for crops not indicated below
Oats	Oats
Rice	Rice
Rye	Rye
Sorghum	Sorghum
Soybeans	Soybean
Sugar Beets	Potatoes and tubers
Sunflowers	Generic value for crops not indicated below
Wheat (Sub-Crop Name Winter Wheat)	Winter wheat
Wheat (Sub-Crop Name Spring Wheat)	Spring wheat
Peanuts	Peanuts
alfalfa	Alfalfa
alfalfa_annual	Alfalfa
alfalfa_grazing	Alfalfa
alkali_grass	Non-N-fixing forages
alsike_clover	N-fixing forages
annual_rye	Rye
annual_ryegrass	Non-N-fixing forages

Table 4: Index Between SEP Crop Type and Grower Survey CropName

Grower Survey CropName	**(SEP-Parameters-v1.1a)**
austrian_winter_peas	Beans & pulses
balansa_clover	N-fixing forages
barley	Barley
bentgrass	Non-N-fixing forages
berseem_clover	N-fixing forages
bluegrass	Non-N-fixing forages
bromegrass	Non-N-fixing forages
buckwheat	Generic value for crops not indicated below
canadian_forage_pearl_millet	Generic value for crops not indicated below
cereal_rye	Rye
chickpea	Beans & pulses
chicory	Generic value for crops not indicated below
cicer_milk_vetch	N-fixing forages
clovers	N-fixing forages
cowpea	Beans & pulses
crimson_clover	N-fixing forages
crown_vetch	N-fixing forages
daikon_radish	Generic value for crops not indicated below
faba_beans	Beans & pulses
fall_rye	Rye
fenugreek	Beans & pulses
fescues	Non-N-fixing forages
field_peas	Beans & pulses
flax	Generic value for crops not indicated below
fodder_rape	Generic value for crops not indicated below
forage_chicory	Generic value for crops not indicated below
forage_peas	Beans & pulses
forage_radish	Generic value for crops not indicated below
forage_rape	Generic value for crops not indicated below
galega	Generic value for crops not indicated below
german_millet	Millet
grazing_turnip	Generic value for crops not indicated below
hairy_vetch	N-fixing forages
hemp	Generic value for crops not indicated below
hemp_sun	Beans & pulses
japanese_millet	Millet
kale	Generic value for crops not indicated below
ladino_clover	N-fixing forages
lawn_and_turf_grasses	Non-N-fixing forages
lentil	Beans & pulses
milk_vetch	N-fixing forages
millet	Millet
mixtures_various	Non-N-fixing forages
mustard	Generic value for crops not indicated below
native_grasses_and_flowers	Non-N-fixing forages
oats	Oats
oilseed_radish	Generic value for crops not indicated below
orchard_grass	Non-N-fixing forages
other	Generic value for crops not indicated below
pearl_millet	Millet
peas	Beans & pulses

Table 4: Index Between SEP Crop Type and Grower Survey CropName

Grower Survey CropName	**(SEP-Parameters-v1.1a)**
perennial_rye_grass	Non-N-fixing forages
persian_clover	N-fixing forages
phacelia	Generic value for crops not indicated below
radish	Generic value for crops not indicated below
rape	Generic value for crops not indicated below
red_clover	N-fixing forages
reed_canary_grass	Non-N-fixing forages
rye	Rye
rye_triticale	Rye
rye_grass_annual	Rye
rye_grass_perennial	Non-N-fixing forages
sorghum	Sorghum
spring_oats	Oats
spring_peas	Beans & pulses
strawberry_clover	N-fixing forages
sudangrass	Sorghum
sugarbeet	Potatoes and tubers
sunflower	Generic value for crops not indicated below
sweet_clover	N-fixing forages
teff_grass	Generic Grains
tillage_turnip	Generic value for crops not indicated below
timothy_grass	Non-N-fixing forages
toro_radish	Generic value for crops not indicated below
trefoil	N-fixing forages
triticale	Generic Grains
turnip	Generic value for crops not indicated below
vetch	N-fixing forages
vetch_fescue	N-fixing forages
wheat_grasses_including_ stream-bank_wild_rye	Generic value for crops not indicated below
wheat_winter_or_spring	Winter wheat
white_clover	N-fixing forages
white_lupins	N-fixing forages
white_mustard	Generic value for crops not indicated below
white_proso_millet	Millet
wildflower_mixtures	Non-N-fixing forages
winter_barley	Barley
winter_peas	Beans & pulses
winter_rye	Rye
winter_triticale	Generic Grains
yellow_white_mustard	Generic value for crops not indicated below

Table 5: Cover crop biomass accumulation by region.

Cover Crop Name (from Ruis et al. 2019)	**Temperature Region**[a]	**Moisture Region**[b]	**Mean Biomass Dry Weight (Mg/Ha)**	**Standard Deviation (Mg/Ha)**
Annual Legume	cold	humid	1.891	1.654
Annual Legume	cold	semiarid	2.119	1.221

Table 5: Cover crop biomass accumulation by region.

Cover Crop Name (from Ruis et al. 2019)	**Temperature Region**[a]	**Moisture Region**[b]	**Mean Biomass Dry Weight (Mg/Ha)**	**Standard Deviation (Mg/Ha)**
Annual Legume	mild	humid	2.975	2.137
Annual Legume	mild	semiarid	2.365	1.565
Annual Legume	warm	humid	5.365	2.669
Annual Legume	warm	semiarid	3.122	2.56
Annual Ryegrass	cold	humid	1.94116	1.56812
Annual Ryegrass	cold	semiarid	0.13528	0.15273
Annual Ryegrass	mild	humid	2.38839	1.80097
Annual Ryegrass	mild	semiarid	–	–
Annual Ryegrass	warm	humid	8.31054	7.22216
Annual Ryegrass	warm	semiarid	2.72707	2.07654
Barley	cold	humid	3.08361	4.42263
Barley	cold	semiarid	–	–
Barley	mild	humid	2.74802	2.44324
Barley	mild	semiarid	1.62639	2.1988
Barley	warm	humid	2.41252	1.95688
Barley	warm	semiarid	5.75447	5.46499
Brassica	cold	humid	3.113	1.697
Brassica	cold	semiarid	1.38	0.861
Brassica	mild	humid	2.19	1.572
Brassica	mild	semiarid	2.387	1.279
Brassica	warm	humid	3.775	1.634
Brassica	warm	semiarid	2.192	1.062
Buckwheat	cold	humid	2.5962	1.45032
Buckwheat	cold	semiarid	–	–
Buckwheat	mild	humid	3.36886	1.05113
Buckwheat	mild	semiarid	2.49318	2.40007
Buckwheat	warm	humid	3.50415	0
Buckwheat	warm	semiarid	–	–
C3 Grass	cold	humid	2.805	2.559
C3 Grass	cold	semiarid	2.263	1.377
C3 Grass	mild	humid	3.545	3.029
C3 Grass	mild	semiarid	2.979	1.929
C3 Grass	warm	humid	4.811	3.212
C3 Grass	warm	semiarid	4.618	3.931
C4 grass	cold	humid	–	–
C4 grass	cold	semiarid	–	–
C4 grass	mild	humid	–	–
C4 grass	mild	semiarid	–	–
C4 grass	warm	humid	–	–
C4 grass	warm	semiarid	–	–
Canola	cold	humid	2.00129	3.16143
Canola	cold	semiarid	0.82897	0.8258
Canola	mild	humid	2.32739	4.09898
Canola	mild	semiarid	–	–
Canola	warm	humid	–	–
Canola	warm	semiarid	–	–
Clover	cold	humid	1.838	1.986
Clover	cold	semiarid	1.001	1.756

Table 5: Cover crop biomass accumulation by region.

Cover Crop Name (from Ruis et al. 2019)	**Temperature Region**[a]	**Moisture Region**[b]	**Mean Biomass Dry Weight (Mg/Ha)**	**Standard Deviation (Mg/Ha)**
Clover	mild	humid	1.791	1.946
Clover	mild	semiarid	2.175	1.007
Clover	warm	humid	4.298	2.119
Clover	warm	semiarid	–	–
Complex Mix	cold	humid	4.98868	7.85334
Complex Mix	cold	semiarid	1.47774	2.27263
Complex Mix	mild	humid	2.91168	5.14568
Complex Mix	mild	semiarid	2.42221	4.14287
Complex Mix	warm	humid	11.4778	7.95426
Complex Mix	warm	semiarid	3.41415	3.41814
Cowpea	cold	humid	0.9942	0.91549
Cowpea	cold	semiarid	–	–
Cowpea	mild	humid	2.90682	2.42468
Cowpea	mild	semiarid	–	–
Cowpea	warm	humid	4.69803	2.30818
Cowpea	warm	semiarid	6.0059	3.07516
Crimson Clover	cold	humid	2.46352	4.44067
Crimson Clover	cold	semiarid	0.25508	0.49434
Crimson Clover	mild	humid	1.94854	3.98153
Crimson Clover	mild	semiarid	–	–
Crimson Clover	warm	humid	4.29784	6.41704
Crimson Clover	warm	semiarid	–	–
Hairy Vetch	cold	humid	1.90651	1.62529
Hairy Vetch	cold	semiarid	1.05736	0.75727
Hairy Vetch	mild	humid	3.09438	2.09866
Hairy Vetch	mild	semiarid	0.42049	-0.00367
Hairy Vetch	warm	humid	4.3892	1.7926
Hairy Vetch	warm	semiarid	2.61217	2.39016
Millet	cold	humid	5.02287	3.48167
Millet	cold	semiarid	1.51837	1.18444
Millet	mild	humid	5.75586	4.19848
Millet	mild	semiarid	–	–
Millet	warm	humid	5.34989	2.36757
Millet	warm	semiarid	–	–
Oat	cold	humid	2.67092	2.38531
Oat	cold	semiarid	2.85374	0.95198
Oat	mild	humid	3.03266	2.23691
Oat	mild	semiarid	3.72019	1.08549
Oat	warm	humid	3.43473	2.46032
Oat	warm	semiarid	4.98484	3.08154
Other	cold	humid	4.125	2.219
Other	cold	semiarid	1.268	1.267
Other	mild	humid	3.379	2.39
Other	mild	semiarid	2.713	1.471
Other	warm	humid	5.472	3.441
Other	warm	semiarid	6.6	1.253
Pea	cold	humid	2.85768	2.75693
Pea	cold	semiarid	2.50813	1.39099

Table 5: Cover crop biomass accumulation by region.

Cover Crop Name (from Ruis et al. 2019)	**Temperature Region**[a]	**Moisture Region**[b]	**Mean Biomass Dry Weight (Mg/Ha)**	**Standard Deviation (Mg/Ha)**
Pea	mild	humid	1.86663	1.39073
Pea	mild	semiarid	2.50361	1.67681
Pea	warm	humid	4.51764	1.45417
Pea	warm	semiarid	3.12973	3.22943
Radish	cold	humid	3.18618	1.85934
Radish	cold	semiarid	1.42034	0.95697
Radish	mild	humid	3.04086	1.85107
Radish	mild	semiarid	2.22372	1.44973
Radish	warm	humid	3.40729	1.09398
Radish	warm	semiarid	3.18304	1.30963
Rapeseed	cold	humid	4.09432	2.17603
Rapeseed	cold	semiarid	0.17871	0.15889
Rapeseed	mild	humid	0.9037	1.13482
Rapeseed	mild	semiarid	2.7463	0.90337
Rapeseed	warm	humid	4.0412	2.46905
Rapeseed	warm	semiarid	1.86175	0.98003
Red Clover	cold	humid	1.61051	1.98933
Red Clover	cold	semiarid	1.113	1.98398
Red Clover	mild	humid	1.64962	1.86814
Red Clover	mild	semiarid	2.17503	1.00739
Red Clover	warm	humid	–	–
Red Clover	warm	semiarid	–	–
Rye	cold	humid	2.87061	3.01834
Rye	cold	semiarid	2.0437	1.7518
Rye	mild	humid	4.63221	3.86331
Rye	mild	semiarid	2.51414	2.39162
Rye	warm	humid	5.40646	3.31456
Rye	warm	semiarid	4.3531	3.93497
Sorghum	cold	humid	4.71292	3.24151
Sorghum	cold	semiarid	–	–
Sorghum	mild	humid	6.10788	3.49666
Sorghum	mild	semiarid	0.66383	0.00661
Sorghum	warm	humid	8.19564	3.81433
Sorghum	warm	semiarid	5.38795	0.00579
Sunn Hemp	cold	humid	1.77981	1.36165
Sunn Hemp	cold	semiarid	–	–
Sunn Hemp	mild	humid	4.05203	3.10759
Sunn Hemp	mild	semiarid	–	–
Sunn Hemp	warm	humid	7.11404	4.36376
Sunn Hemp	warm	semiarid	–	–
Triticale	cold	humid	2.85208	4.9562
Triticale	cold	semiarid	1.50794	3.04678
Triticale	mild	humid	1.46952	3.9985
Triticale	mild	semiarid	3.48395	5.57131
Triticale	warm	humid	4.63993	8.61686
Triticale	warm	semiarid	–	–
Turnip	cold	humid	4.42372	0.2892
Turnip	cold	semiarid	2.15931	0.97781

Table 5: Cover crop biomass accumulation by region.

Cover Crop Name (from Ruis et al. 2019)	**Temperature Region**[a]	**Moisture Region**[b]	**Mean Biomass Dry Weight (Mg/Ha)**	**Standard Deviation (Mg/Ha)**
Turnip	mild	humid	2.36035	1.0024
Turnip	mild	semiarid	–	–
Turnip	warm	humid	5.99007	0
Turnip	warm	semiarid	–	–
Two-Species Mix	cold	humid	3.64808	1.73663
Two-Species Mix	cold	semiarid	0.95434	1.9742
Two-Species Mix	mild	humid	3.9389	2.75189
Two-Species Mix	mild	semiarid	3.32549	0.76137
Two-Species Mix	warm	humid	5.40568	3.61186
Two-Species Mix	warm	semiarid	7.39588	1.56567
Wheat	cold	humid	–	–
Wheat	cold	semiarid	3.30641	1.90515
Wheat	mild	humid	2.77782	2.42691
Wheat	mild	semiarid	2.58343	2.14574
Wheat	warm	humid	3.4056	2.07988
Wheat	warm	semiarid	–	–

- ([a]). Temperature regime is determined by 30-year weather time series are used to compute USDA Hardiness Zones:
  - If USDA Hardiness Zone > 7 then temperature region = warm
  - If USDA Hardiness Zone = 5, 6, or 7 then temperature region = mild
  - If USDA Hardiness Zone < 5 then temperature region = cold
- ([b]). Moisture regime is determined by 30-year precipitation averages:
  - If mean natural precipitation is greater than 750mm or irrigation occurs on the field, then the moisture region = humid
  - If mean natural precipitation is less than or equal to 750mm and there is no irrigation, then the moisture region = semiarid

CropName	**Cover Crop Name (from Ruis et al. 2019)**
alfalfa	Clover
alfalfa_annual	Annual Legume
alfalfa_grazing	Clover
alkali_grass	C3 Grass
alsike_clover	Clover
annual_rye	Rye
annual_ryegrass	Annual Ryegrass
austrian_winter_peas	Pea
balansa_clover	Clover
barley	Barley
bentgrass	C3 Grass
berseem_clover	Clover
bluegrass	C3 Grass
bromegrass	C3 Grass
buckwheat	Buckwheat

CropName	Cover Crop Name (from Ruis et al. 2019)
canadian_forage_pearl_millet	Millet
cereal_rye	Rye
chickpea	Pea
chicory	Other
cicer_milk_vetch	Hairy Vetch
clovers	Clover
cowpea	Cowpea
crimson_clover	Crimson Clover
crown_vetch	Hairy Vetch
daikon_radish	Radish
faba_beans	Annual Legume
fall_rye	Rye
fenugreek	Annual Legume
fescues	C3 Grass
field_peas	Pea
flax	Other
fodder_rape	Rapeseed
forage_chicory	Other
forage_peas	Pea
forage_radish	Radish
forage_rape	Rapeseed
galega	Annual Legume
german_millet	Millet
grazing_turnip	Turnip
hairy_vetch	Hairy Vetch
hemp	Other
hemp_sun	Sunn Hemp
japanese_millet	Millet
kale	Brassica
ladino_clover	Clover
lawn_and_turf_grasses	C4 grass
lentil	Annual Legume
milk_vetch	Hairy Vetch
millet	Millet
mixtures_various	Two-Species Mix
mustard	Brassica
native_grasses_and_flowers	Other
oats	Oat
oilseed_radish	Radish
orchard_grass	C3 Grass
other	Other
pearl_millet	Millet
peas	Pea
perennial_rye_grass	C3 Grass
persian_clover	Clover
phacelia	Other
radish	Radish
rape	Rapeseed
red_clover	Red Clover
reed_canary_grass	C3 Grass
rye	Rye
rye_triticale	Rye

CropName	Cover Crop Name (from Ruis et al. 2019)
rye_grass_annual	Annual Ryegrass
rye_grass_perennial	C3 Grass
sorghum	Sorghum
spring_oats	Oat
spring_peas	Pea
strawberry_clover	Clover
sudangrass	Sorghum
sugarbeet	Other
sunflower	Other
sweet_clover	Clover
teff_grass	C3 Grass
tillage_turnip	Turnip
timothy_grass	C3 Grass
toro_radish	Radish
trefoil	Clover
triticale	Rye
turnip	Turnip
vetch	Hairy Vetch
vetch_fescue	C3 Grass
wheat_grasses_including_streambank_wild_rye	Wheat
wheat_winter_or_spring	Wheat
white_clover	Clover
white_lupins	Annual Legume
white_mustard	Brassica
white_proso_millet	Millet
wildflower_mixtures	Other
winter_barley	Barley
winter_peas	Pea
winter_rye	Rye
winter_triticale	Rye
yellow_white_mustard	Brassica

- Fraction of N in dry matter for plant species ($N_content, g$) is provided by `SEP-Parameters-v-1.1a`. The aboveground and below ground Ncontent values (N AG(T)) and (N BG (T)), respectively, are used. CropName comes from the Grower Survey and is mapped to SEP crops according to Table 2.
- See section on Equation 5.20 for details on calculating EF_{Ndirect}
- Molar mass ratio of N_2O to N and GWP_{N2O} are both provided by the SEP (Table 5.1 and Equation 5.25).

3.6 Equation 5.27. Nitrous Oxide Emissions from Biomass Burning

- See details on Equation 5.15 for calculation of $\text{MB}_{c,s,t}$
- See details on Equation 5.15 for calculation of CF_c
- The methane emission factor for the burning of agricultural residue (EF_c, N_2O) is taken from the "Biomass Burning tab of `SEP-Parameters-v-1.1a` and is the same for all crops (0.07 gN_2O/kg dry matter burnt).
- GWP_{N2O} and conversion from kg to ton are provided by the SEP (Table 5.1 and Equation 5.27).

4 Carbon Dioxide Emissions

4.1 Equation 5.28. Net Change in Carbon Dioxide Emission Reductions From Fossil Fuel.

- See details on Equation 5.29b for calculation of average carbon dioxide emissions deductions from fossil fuel use

4.2 Equation 5.29b. Carbon Dioxide emissions from fossil fuels based on agronomic practice.

- See the "Proposal for an alternative approach for estimating CO_2 emissions from fossil fuel use in the Soil Enrichment Protocol" for equation specifics of 5.29b
  - $\overline{FFC_{e,s,t}}$ is the average consumption of diesel fuel per unit area for stratum s in an agronomic practice e in cultivation cycle t [gal/acre].
    - Agronomic practices (e) are from field operation data submitted by the grower or gapfilled following the protocols specified in `IndigoCarbon_US-1_2022_0031`.
    - Fuel use values are taken directly from Table 5, with the exceptions of derived values and proxy matches outlined in the following mapping logic section.
  - $EF_{\mathrm{CO_2}}$ is the emission factor for diesel fuel (10,180 grams of CO_2/gallon of diesel)
    - The emission factor for diesel fuel (EF_{CO2}) is from the EPA Greenhouse Gas Equivalencies (EPA, 2022).

4.3 Mapping logic (implementation in Table 5)

- The per acre diesel fuel use of each agronomic practice e is from the Proposal for an alternative approach for estimating CO_2 emissions from fossil fuel use in the Soil Enrichment Protocol (Revised) (Fossil Fuel Proposal) and is mapped to Indigo's internal agronomy data service (ADS) practices, methods, and equipment (Table 5).
- Mapping of diesel fuel use were completed as follows. If:
  - ADS equipment can be mapped to an operation in the Fossil Fuel Proposal.
    - Fossil Fuel Proposal diesel consumption value directly assigned to ADS equipment.
  - ADS equipment can be mapped to several operations in the Fossil Fuel Proposal.
    - Matching diesel consumption values in the Fossil Fuel Proposal were averaged.
  - ADS equipment cannot be mapped to the Fossil Fuel Proposal but a fuel use value can be derived.
    - A simple training model was created: The cultivation factor from the DayCent Manual (Hartman et al., 2020, Table 15) was regressed with diesel fuel consumption from the Fossil Fuel Proposal for all direct ADS- Fossil Fuel Proposal matches.
    - The resultant linear equation was used to derive diesel fuel consumption (dependent variable) from the cultivation factor (independent variable) for any non-direct-matched ADS method.
    - Diesel fuel consumption derived in this manner is labeled "predicted" in Table 5.
  - ADS equipment cannot be mapped and a value cannot be predicted by the above method.
    - A proxy match was made between the Fossil Fuel Proposal and ADS equipment

4.4 Additional mapping rules (examples in Table 5):

- Special rule for planting
  - The planting equipment was matched based on the crop type and whether the modeled crop was a cover crop
  - Any logic to discern planting equipment based on tillage status was not included. This is because:

        * The crop type mapping of planting equipment is more representative than the generalized "no-till planter". Additionally, not all growers use no-till planters under no-till.

- Special rule for harvest
    - The harvest equipment was matched based on the crop type and harvest method (Hay, Silage, Grain only, etc.) for a harvest event.
        * For example, soybean grain would be harvested with a combine for beans and corn by a combine for corn/sorghum from the Fossil Fuel Proposal
    - Unlike tillage, planting, or application events, harvests are often not a discrete operation. The grain must be harvested and hauled. In the case of hay, it needs to be cut, raked, baled, and hauled.
        * The diesel fuel (gal/acre) of every crop-specific equipment used during a harvest operation was summed for each case.
- Special rule for termination
    - The most conservative, and most representative, termination option of a rotatory mower was chosen for a mow-termination of cover crop.
    - If a tillage pass was not used to terminate a cover crop and information is not collected on non-fertilizer chemical applications, include one additional sprayer operation on the date of cover crop termination.
    - Cover crops with termination type "left in field" had no associated diesel fuel use.
    - If a cover crop failed, it was treated as "left in field".
- Special rule for NULL occurrences
    - If there is an occurrence whereby the ADS_equpiment is not populated, the most conservative equipment from the relevant method is added.
- Special rules for chemical application during tillage reduction
    - If data are not collected on non-fertilizer chemical application events, include one additional sprayer operation in the project condition when tillage is reduced between the baseline and project conditions.

Table 7: Fuel use mapping by tillage operation.

Category	ADS Method	ADS Crop Name	Cover Crop	ADS Equipment	Fossil Fuel Proposal	Gallons of Diesel per Acre
tillage	MOLDBOARD	n/a	n/a	MOLDBOARD	Plow 8 inches deep	1.68
tillage	MOLDBOARD	n/a	n/a	HEAVY_OFFSET_ DISK	Heavy offset disk	0.95
tillage	MOLDBOARD	n/a	n/a	NULL	Plow 8 inches deep	1.68
tillage	DEEP_RIPPER	n/a	n/a	DISK_RIPPER	predicted	1.11
tillage	DEEP_RIPPER	n/a	n/a	INLINE_RIPPER	predicted	1.11
tillage	DEEP_RIPPER	n/a	n/a	V_RIPPER	predicted	1.11
tillage	DEEP_RIPPER	n/a	n/a	NULL	predicted	1.11
tillage	DISK	n/a	n/a	DISK	predicted	0.74
tillage	DISK	n/a	n/a	VERTICAL_DISK	predicted	0.61
tillage	DISK	n/a	n/a	DISK_AND_ROLLER	predicted	0.74
tillage	DISK	n/a	n/a	TANDEM_DISK_ HARROW	Averaged: [Tandem disk, stalks | Tandem disk, chiseled | Tandem disk, plowed]	0.55
tillage	DISK	n/a	n/a	NULL	predicted	0.74
tillage	CHISEL	n/a	n/a	CHISEL_PLOW	Chisel plow	1.1
tillage	CHISEL	n/a	n/a	SWEEP_PLOW	Sweep plow	0.6
tillage	CHISEL	n/a	n/a	SWEEP_PLOW_ WITH_CULTIPACKER	Sweep plow	0.6
tillage	CHISEL	n/a	n/a	NULL	Chisel plow	1.1
tillage	RIDGE_TILL	n/a	n/a	RIDGE_CULTIVATOR	predicted	0.39
tillage	RIDGE_TILL	n/a	n/a	LISTER	predicted	0.56
tillage	RIDGE_TILL	n/a	n/a	NULL	predicted	0.56
tillage	STRIP_TILL	n/a	n/a	STRIP_TILLER	predicted	1.09
tillage	STRIP_TILL	n/a	n/a	NULL	predicted	1.09
tillage	VERTICAL_TILL	n/a	n/a	VERTICAL_TILLER	predicted	0.61
tillage	VERTICAL_TILL	n/a	n/a	NULL	predicted	0.61
tillage	ROW_ CULTIVATOR	n/a	n/a	ROW_CULTIVATOR	Cultivate row crops	0.45
tillage	ROW_ CULTIVATOR	n/a	n/a	ROTARY_HOE	Rotary hoe	0.25
tillage	ROW_ CULTIVATOR	n/a	n/a	NULL	Cultivate row crops	0.45
tillage	FINISHER	n/a	n/a	CULTIPACKER_ PACKER	predicted	0.39
tillage	FINISHER	n/a	n/a	FIELD_CULTIVATOR	Field cultivate	0.6
tillage	FINISHER	n/a	n/a	POWER_TILLER	predicted	0.29
tillage	FINISHER	n/a	n/a	POWER_HARROW	predicted	0.54
tillage	FINISHER	n/a	n/a	BEDDER	predicted	1.22

Table 7: Fuel use mapping by tillage operation.

Category	**ADS Method**	**ADS Crop Name**	**Cover Crop**	**ADS Equipment**	**Fossil Fuel Proposal**	**Gallons of Diesel per Acre**
tillage	FINISHER	n/a	n/a	ROLLING_BASKET	predicted	0.49
tillage	FINISHER	n/a	n/a	SOIL_FINISHER	predicted	0.25
tillage	FINISHER	n/a	n/a	HARROW	predicted	0.43
tillage	FINISHER	n/a	n/a	DIAMOND_ HARROW	predicted	0.79
tillage	FINISHER	n/a	n/a	NULL	predicted	1.22
planting		cover_crop_generic OR [Alfalfa, Canola, Oats, Rye, Barley, Beans/Peas, Flax, Triticale, Wheat]	TRUE		Grain drill	0.35
planting		[Corn, Sorghum, Soybeans, Cotton, Sunflowers, Peanuts, Sugar Beets]	TRUE		Planting row crops	0.5
planting		[Corn, Sorghum, Soybeans, Cotton, Sunflowers, Peanuts, Sugar Beets]	FALSE		Planting row crops	0.5
planting		[Alfalfa, Canola, Oats, Rye, Barley, Beans/Peas, Flax, Triticale, Rice, Wheat]	FALSE		Grain drill	0.35
harvest	BALED_STRAW	cover_crop_generic OR [Alfalfa, Barley, Corn, Oats, Rye, Sorghum, Soybeans, Triticale, Wheat]	n/a		Baler | Haul small grains	1.05
harvest	HAY	cover_crop_generic OR [Alfalfa, Barley, Oats, Rye, Soybeans, Tiriticale, Wheat]	n/a		Mower (cutterbar) | Rake, single | Baler | Haul small grains	1.9
harvest	HAY	[Corn, Sorghum]	n/a		Forage harvester, corn silage | Rake, single | Baler | Haul small grains	4.9

Table 7: Fuel use mapping by tillage operation.

Category	ADS Method	ADS Crop Name	Cover Crop	ADS Equipment	Fossil Fuel Proposal	Gallons of Diesel per Acre
harvest	HAY	NULL	n/a		Forage harvester, corn silage | Rake, single | Baler | Haul small grains	4.9
harvest	HAYLAGE	cover_crop_generic OR [Alfalfa, Oats, Wheat]	n/a		Forage harvester, haylage | Rake, single | Baler | Haul small grains	2.55
harvest	HAYLAGE	[Corn, Sorghum]	n/a		Forage harvester, corn silage | Rake, single | Baler | Haul small grains	4.9
harvest	HAYLAGE	NULL	n/a		Forage harvester, corn silage | Rake, single | Baler | Haul small grains	4.9
harvest	SILAGE	[Corn, Sorghum]	n/a		Forage harvester, corn silage | Haul small grains	4.2
harvest	SILAGE	cover_crop_generic OR [Alfalfa, Oats, Soybeans, Wheat]	n/a		Forage harvester, haylage | Haul small grains	1.85
harvest	SILAGE	NULL	n/a		Forage harvester, corn silage | Haul small grains	4.2
harvest	GRAIN_ONLY	cover_crop_generic OR [Canola, Rice, Oats, Rye, Barley, Flax, Wheat, Triticale]	n/a		Combine, small grains | Haul small grains	1.6
harvest	GRAIN_ONLY	[Beans/Peas, Soybeans, Sunflowers]	n/a		Combine, beans | Haul small grains	1.7
harvest	GRAIN_ONLY	[Corn, Sorghum]	n/a		Combine, corn and grain sorghum | Haul small grains	2.2

Table 7: Fuel use mapping by tillage operation.

Category	ADS Method	ADS Crop Name	Cover Crop	ADS Equipment	Fossil Fuel Proposal	Gallons of Diesel per Acre
harvest	GRAIN_ONLY	NULL	n/a		Combine, corn and grain sorghum | Haul small grains	2.2
harvest	GRAIN_WITH_ RESIDUE_REMOVAL	cover_crop_generic OR [Canola, Rice, Oats, Rye, Barley, Flax, Wheat, Triticale]	n/a		Combine, small grains | Haul small grains | Baler | Haul small grains	2.65
harvest	GRAIN_WITH_ RESIDUE_REMOVAL	[Beans/Peas, Soybeans, Sunflowers]	n/a		Combine, beans | Haul small grains | Baler | Haul small grains	2.75
harvest	GRAIN_WITH_ RESIDUE_REMOVAL	[Corn, Sorghum]	n/a		Combine, corn and grain sorghum | Haul small grains | Baler | Haul small grains	3.25
harvest	GRAIN_WITH_ RESIDUE_REMOVAL	NULL	n/a		Combine, corn and grain sorghum | Haul small grains | Baler | Haul small grains	3.25
harvest	GRAIN_ONLY	[Cotton]	n/a		Averaged: [Combine, small grains | Combine, beans | Combine, corn and grain sorghum] | Haul small grains	1.83
harvest	COTTON_PICKER_ OR_STRIPPER	[Cotton]	n/a		Averaged: [Combine, small grains | Combine, beans | Combine, corn and grain sorghum] | Haul small grains	1.83
harvest	DIGGING	[Sugar Beets, Peanuts]	n/a		Forage harvester, corn silage | Haul small grains	4.2
application	SURFACE	n/a	n/a		Sprayer	0.1

Table 7: Fuel use mapping by tillage operation.

Category	**ADS Method**	**ADS Crop Name**	**Cover Crop**	**ADS Equipment**	**Fossil Fuel Proposal**	**Gallons of Diesel per Acre**
application	INJECTION_OR_ BANDED_0_TO_4_IN	n/a	n/a		predicted	0.47
application	INJECTION_5_ IN_PLUS	n/a	n/a		predicted	0.47
application	INCORPORATION_0_ TO_4_IN	n/a	n/a		Sprayer	0.1
application	INCORPORATION_ 5_TO_12_IN	n/a	n/a		Sprayer	0.1
application	INCORPORATION_ 3_IN_PLUS	n/a	n/a		Sprayer	0.1
application	OTHER					
termination	herbicide	n/a	n/a		Sprayer	0.1
termination	roller_crimper	n/a	n/a		predicted (CULTI-PACKER_PACKER)	0.39
termination	mowing	n/a	n/a		Rotary mower	0.8
termination	winter_kill	n/a	n/a			0
termination	tillage	n/a	n/a			0

References

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5 Summary of Changes

Section	Date	Summary of Changes
Eqn 5.11b	June 24, 2024	Updated logic to calculate $AGD_{l,s,t}$; added source for $B_{O,I}$ and VS_l.
Eqn 5.20	June 24, 2024	Clarifying language changes and updates; Added reasoning for GWP values used.
Eqn 5.24	June 24, 2024	Updated value for $Frac_{LEACHMD}$ (0.24 in wet climates).
Eqn 5.29b	June 24, 2024	Updated text to reflect fossil fuel accounting.
Eqn 5.20	June 24, 2024	Updated to reflect use of IPCC data inputs.